

Rahul Kothagundla

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Professional Summary

Software Engineer with 1.5+ years building distributed systems at scale. Delivered 70% performance improvements, maintained data integrity across 50K+ daily records and earned 5-star rating. Expertise in backend development, real-time data processing (Kafka, Spark) and code quality. Seeking SWE role at Google Payments to apply distributed systems expertise to payment infrastructure serving billions.

Technical Skills

Languages:	Python, Java, C++, SQL, C
Frameworks & Libraries:	Apache Kafka, Apache Spark, FastAPI, PySpark, Docker, REST APIs
Databases & Storage:	PostgreSQL, Redis, SQL Server
Cloud & DevOps:	Azure, AWS, Databricks, CI/CD, Git, Docker Compose
Core Competencies:	Data Structures & Algorithms, Object-Oriented Design, System Design, Distributed Systems

Professional Experience

Tata Consultancy Services <i>Systems Engineer</i>	Jun 2024 – Present <i>Hyderabad</i>
- Engineered distributed data processing system using Azure Databricks and Python, re-architecting computational layer to offload heavy operations, reducing reload time by 70% (15min → 7min). - Designed production-grade self-monitoring pipeline with automated error detection and failure handling, maintaining data integrity across 50K+ daily records. - Built end-to-end ETL pipelines from scratch implementing multi-stage validation and transformation logic, enabling real-time business intelligence for executive financial decision-making. - Authored comprehensive technical documentation covering architecture and debugging procedures, reducing engineer ramp-up from 7 weeks to 4 days (90% improvement). Earned 5-star performance rating.	
Atomstate Technologies <i>Machine Learning Engineer</i>	Jun 2023 – Sep 2023 <i>Hyderabad</i>
- Built and deployed sentiment analysis engine using NLP, processing unstructured text at scale with automated pipeline for data ingestion, model inference and storage. - Developed predictive models processing large datasets for election analysis platform, generating strategic insights for campaign resource allocation.	
Zummit Infolabs <i>Junior Data Scientist</i>	Nov 2022 – Mar 2023 <i>Remote</i>
Applied NLP techniques to extract insights from unstructured data, creating analytical models converting raw text into structured business intelligence.	

Technical Projects

Distributed Payment Transaction System Python, FastAPI, PostgreSQL, Redis, Kafka	
- Built production-grade payment processing API with idempotency keys, ACID transactions and fraud detection, demonstrating expertise in payment systems architecture. - Implemented distributed locking with Redis, webhook notifications, transaction state machine and comprehensive test coverage (85%+ using pytest). - Containerized with Docker Compose, deployed with health monitoring architecture ready for billion-user scale.	
Real-Time Stock Market Streaming Python, Kafka, Spark, PostgreSQL, Redis	
- Built distributed streaming platform processing live market data with sub-30s latency, handling 1000+ events/second with fault-tolerant design. - Implemented Kafka (3 partitions), Spark Structured Streaming with exactly-once semantics, PostgreSQL with composite indexes and Redis caching (60% load reduction). - Achieved 99.9% uptime with containerized microservices and auto-recovery mechanisms.	

Education

B.Tech in Computer Science And Engineering – Data Science VNR VJIET	2020 – 2024 CGPA: 8.55/10
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Certifications & Problem Solving

AWS Cloud Quest: Generative AI Practitioner (Jan 2026) | Amazon ML Summer School 2023
HackerRank Problem Solving | ISRO Geo Data Processing with Python